

Polynomial Regression: Predictive Modeling for Geothermal Expansion

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<https://github.com/dhyanspatel49/Assignment-Polynomial-Regression>

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Abstract

This report develops polynomial regression models to optimize a multi-stage geothermal power plant (Phase 1) and to map a subterranean thermal reservoir (Phase 2). For each dataset, polynomial feature expansion, feature standardization and two regularized estimators, Ridge (L_2) and Lasso (L_1), were compared across polynomial degrees using 5-fold cross-validation. The best model was chosen per problem by mean cross-validated R^2 . Lasso at degree 5 was selected for Phase 1 (CV $R^2 = 0.9682$, MSE = 0.3283), consistent with a response that depends on a sparse subset of polynomial terms. Ridge at degree 10 was selected for Phase 2 (CV $R^2 = 0.9937$, MSE = 0.2601), consistent with a smooth polynomial field in which most terms carry signal.

1 Introduction

The objective is to predict a continuous target y from a set of input variables for two problems, using polynomial regression exclusively. Polynomial expansion produces many strongly correlated features, so ordinary least squares (OLS) is unstable here and a regularized approach is required. Rather than assume one regularizer suits both problems, Ridge and Lasso were evaluated side by side on each dataset. They encode different assumptions about the underlying function, and the data decided which one fits better.

2 Methodology and Techniques

The same pipeline was applied to both problems: `PolynomialFeatures` $\rightarrow$ `StandardScaler` $\rightarrow$ regularized linear regressor (Ridge or Lasso), implemented in scikit-learn.

2.1 Polynomial Feature Expansion

Polynomial expansion generates every monomial whose total power is at most the chosen degree d . For m inputs this yields $\binom{m+d}{d} - 1$ features, which grows rapidly. For example, 6 inputs at degree 5 give 461 features and at degree 8 give 3,002, while 3 inputs at degree 10 give 285 and at degree 20 give 1,770.

2.2 Feature Scaling

Polynomial terms differ greatly in magnitude, and penalized regression is sensitive to feature scale. A `StandardScaler` applied after expansion gives every polynomial feature zero mean and unit variance, so the penalty treats all terms equally and numerical overflow at high degrees is avoided.

2.3 Why Regularization Is Needed

With thousands of correlated monomials, $X^T X$ is close to singular and OLS coefficients become unstable and overfit. Both regularizers add a penalty to the residual sum of squares (RSS):

$$\text{Ridge: } \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p \beta_j^2, \quad (1)$$

$$\text{Lasso: } \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p |\beta_j|. \quad (2)$$

The two penalties behave differently:

- **Ridge** (L_2) shrinks all coefficients smoothly toward zero but never sets them exactly to zero. It suits problems where many terms contribute, and it spreads weight across correlated terms.
- **Lasso** (L_1) drives the coefficients of uninformative terms exactly to zero, which performs automatic feature selection. It suits problems where only a few terms matter.

Since the true polynomial structure is unknown for each dataset, neither assumption can be fixed in advance, which is why both were compared.

2.4 Model and Degree Selection

The penalty strength α was tuned inside every training fold: **RidgeCV** used a log-spaced grid of 10 values from 10^{-3} to 10^3 (chosen by its built-in efficient leave-one-out cross-validation), and **LassoCV** used 50 automatically generated values with 3-fold internal validation. The degree was then selected with 5-fold cross-validation (**KFold**, shuffled, seed 42) on the training set only, so no test data was used. For each model and degree, the mean validation R^2 and MSE were recorded. The final model is the (estimator, degree) pair with the highest mean R^2 across both estimators, which is then refit on the full training set to predict the test set. Because the best of many candidates is reported, the quoted CV scores are slightly optimistic estimates of test performance.

3 Phase 1: Power Plant Steam Turbine Optimization (var1)

3.1 Context and Approach

Phase 1 predicts the Net Power Score (y) from six turbine parameters ($x_1, \dots, x_6$). The assignment indicates a degree of up to 10. Ridge was therefore evaluated up to degree 10, and its score fell steadily after degree 5 (degree 9: $R^2 = 0.9102$; degree 10: $R^2 = 0.8938$; see Table 1). Lasso is far more expensive at high degree, because its cost grows with the number of features (3,002 at degree 8, and 8,007 at degree 10) and each candidate is refit inside nested cross-validation. Lasso was therefore evaluated up to **degree 8**. This is sufficient: the score of both models peaks at degree 5 and then declines with each additional degree, which is the signature of overfitting. Since the trend is already clearly downward, higher degrees would only add cost without changing the selected model.

3.2 Results

Table 1: Cross-validation metrics for Phase 1 (var1). Best results are in bold. Lasso was not evaluated beyond degree 8 (dashes).

Degree	Ridge		Lasso	
	Mean R^2	Mean MSE	Mean R^2	Mean MSE
1	0.0773	9.6362	0.0750	9.6613
2	0.7122	2.9597	0.7135	2.9543
3	0.9009	1.0147	0.9011	1.0138
4	0.9307	0.7094	0.9398	0.6157
5	0.9512	0.4965	0.9682	0.3283
6	0.9426	0.5886	0.9666	0.3430
7	0.9339	0.6759	0.9643	0.3667
8	0.9173	0.8510	0.9636	0.3736
9	0.9102	0.9236	—	—
10	0.8938	1.0986	—	—

Both models are nearly identical at degrees 1–3, where the polynomial is too simple and the error is dominated by bias. From degree 4 onward Lasso pulls clearly ahead. At degree 5, Lasso’s mean MSE is about 34% lower than Ridge’s (0.3283 versus 0.4965). Ridge degrades quickly beyond degree 5 (R^2 drops from 0.9512 to 0.9173 by degree 8), whereas Lasso degrades only slightly (0.9682 to 0.9636).

Robustness to the test distribution. The test inputs are more extreme than the training inputs: about 49% of test values lie at ± 1 , against about 31% in training (roughly 3.0 versus 1.9 clipped coordinates per sample). Random-fold CV may therefore understate the test error of every model. As a check using training data only, each model was fit on the 75% of training points with the smallest input norm and evaluated on the remaining 25% with the largest norm. Lasso at degree 5 again performed best (MSE $\approx$ 0.51), ahead of Lasso at degree 6 (0.65) and Ridge, whose best result was 1.31 (degree 4). This supports the selected model under a shifted input distribution.

3.3 Why Lasso Succeeded in Phase 1

A 6-variable polynomial of degree 5 contains 461 terms, but there are only 1,000 training samples, and most of these terms (many high-order cross-interactions between the six turbine parameters) are unlikely to influence the output. Lasso exploited this: at degree 5 the selected model kept only **115 of the 461 terms** non-zero. Setting the irrelevant terms exactly to zero removes a large source of variance. Ridge cannot do this, because it keeps all 461 terms with small but non-zero weights, and the accumulated noise from the irrelevant terms lowers its accuracy. The input data also contains many boundary values (about 30% of each feature is clipped at ± 1), which makes the noise-fitting problem worse for a model that retains every term. The same reasoning explains why Lasso degrades much more gracefully at higher degrees: as the degree grows, most newly added terms are irrelevant, and Lasso keeps ignoring them while Ridge keeps fitting them.

Conclusion for Phase 1: Lasso with degree 5 was selected (CV $R^2 = 0.9682$, CV MSE = 0.3283). Final predictions were saved to `BT2024075_pred_var1.csv`.

4 Phase 2: Subterranean Thermal Reservoir Mapping (var2)

4.1 Context and Approach

Phase 2 maps the Thermal Anomaly Score (y) from three spatial coordinates (x_1, x_2, x_3). The heat map is described as a high-degree polynomial (up to degree 20). Because there are only three inputs, even degree 20 gives just 1,770 features, so the full range from 1 to 20 was evaluated for both estimators.

4.2 Results

Table 2: Cross-validation metrics for Phase 2 (var2), selected degrees. Best results are in bold.

Degree	Ridge		Lasso	
	Mean R^2	Mean MSE	Mean R^2	Mean MSE
1	0.1982	32.9530	0.1962	33.0370
2	0.5092	20.4104	0.5093	20.4382
3	0.7226	11.2399	0.7233	11.2098
5	0.9653	1.3973	0.9652	1.4191
7	0.9914	0.3508	0.9872	0.5144
8	0.9932	0.2797	0.9907	0.3757
9	0.9935	0.2675	0.9915	0.3452
10	0.9937	0.2601	0.9921	0.3216
11	0.9936	0.2641	0.9922	0.3175
13	0.9936	0.2654	0.9924	0.3143
15	0.9931	0.2871	0.9925	0.3100
20	0.9921	0.3287	0.9922	0.3209

Both models learn the underlying structure at similar speeds up to degree 5 and then separate. Ridge reaches its best score at degree 10 and holds a stable plateau through degree 14 (Figure 1). Lasso needs more degrees to catch up (best at degree 15) and never reaches Ridge’s accuracy: its best MSE of 0.3100 is about 19% higher than Ridge’s best of 0.2601. Beyond the optimum, the performance of both models declines only gently, since the penalty prevents the coefficient blow-up that unregularized high-degree fits suffer.

4.3 Why Ridge Succeeded in Phase 2

The independence of the coordinates x_1, x_2, x_3 is not a reason to prefer or avoid either estimator. Regularization acts on the polynomial features, and terms such as x_1 , x_1^2 , x_1^3 and x_1x_2 are strongly correlated with one another regardless of how the original coordinates are distributed. What matters is the structure of the target. A subsurface heat map built from a high-degree polynomial in only three variables plausibly has many genuinely non-zero coefficients, so there is little irrelevant structure for Lasso to remove. Two properties of Lasso then work against it:

- **Shrinkage bias on real terms.** The L_1 penalty zeroes out small but real terms and shrinks the large ones by a constant amount, introducing bias when most coefficients matter.
- **Arbitrary choice among correlated terms.** When several correlated monomials explain the same pattern, Lasso tends to keep one and discard the others, which makes it less stable. Ridge shares the weight across them, giving smoother and more reliable predictions.

Ridge shrinks every coefficient proportionally and keeps all terms, which fits this setting. With 1,000 samples and only 285 features at degree 10, the data also comfortably supports estimating all coefficients.

Conclusion for Phase 2: Ridge with degree 10 was selected ($CV R^2 = 0.9937$, $CV MSE = 0.2601$). Final predictions were saved to `BT2024075_pred_var2.csv`.

5 Model Comparison and Discussion

Figure 1 summarizes the cross-validated R^2 of both estimators across degrees for the two problems.

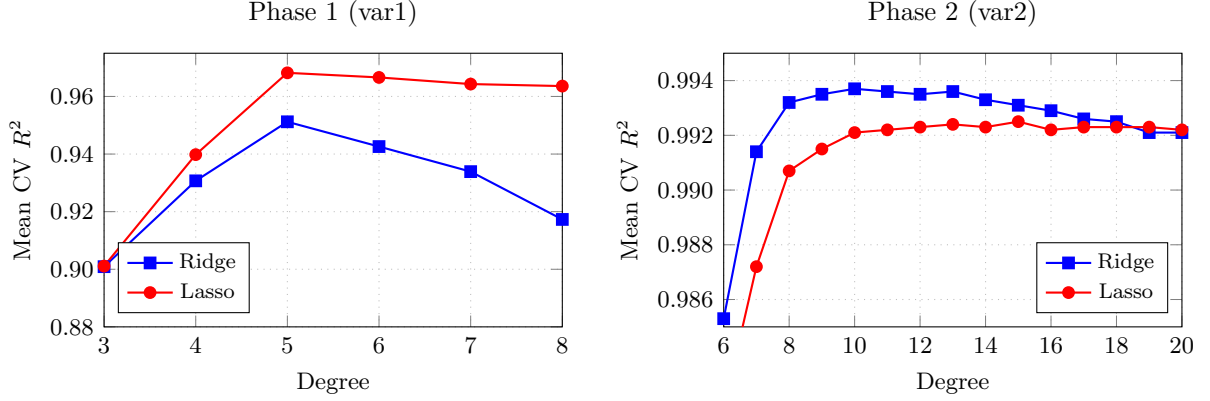


Figure 1: Cross-validated R^2 versus polynomial degree for Ridge and Lasso. Low degrees are omitted so the differences between the models are visible.

Table 3: Selected final models.

Problem	Estimator	Degree	CV R^2	CV MSE
Phase 1 (var1)	Lasso	5	0.9682	0.3283
Phase 2 (var2)	Ridge	10	0.9937	0.2601

The two problems favor opposite regularizers, and the reason appears to be the structure of the target rather than the number of variables or any property of the inputs. Phase 1 seems to have many candidate interaction terms of which only a minority matter (115 of 461 coefficients are non-zero in the selected Lasso model), so selection (Lasso) wins. Phase 2 behaves like a dense and smooth polynomial field in which most terms matter, so shrinkage without selection (Ridge) wins. These are interpretations of the CV results rather than directly verified properties of the data-generating function. In both cases the CV curve rises, peaks and then slowly declines with degree, confirming that the selected degrees balance underfitting and overfitting. All selection used training data only, and the penalty strength was tuned inside each fold.

6 Conclusion

Polynomial regression with feature standardization and regularization, combined with cross-validated model and degree selection, produced accurate models for both problems. Comparing Ridge and Lasso, rather than fixing one in advance, improved Phase 1 (MSE reduced by about 34% over Ridge) and confirmed Ridge as the right choice for Phase 2. The final models are **Lasso (degree 5)** for the turbine optimization and **Ridge (degree 10)** for the thermal reservoir mapping. The code, prediction files and this report are available in the linked GitHub repository.